جامعة الملك عبدالله
للعلوم والتقنية
King Abdullah University of
Science and Technology

MACHINE LEARNING

– CS229 –

# Trial Exam

**Total: 100 Points**

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*Student name:* _____

*Score:* _____

Spring 2025

# Problem 1: Polynomial Regression with Regularization (18 points)

**Statement.** We have four data points in 1D:

$$(x_1, t_1) = (0, 0), \quad (x_2, t_2) = (1,\ 0.8), \quad (x_3, t_3) = (2,\ 1.6), \quad (x_4, t_4) = (3,\ 3.7).$$

We fit the polynomial

$$y(x) \ = \ w_0 \ + \ w_1\, x \ + \ w_2\, x^2,$$

subject to a ridge (L2) penalty $\lambda = 1$:

$$\sum_{n=1}^{4} \big[y(x_n) - t_n\big]^2 \ + \ \lambda \sum_{j=0}^{2} w_j^2.$$

(a) Write the design matrix $\mathbf{\Phi}$.

(b) Formulate the ridge equation.

(c) Solve for $(w_0, w_1, w_2)$ *numerically*, by explicitly writing and inverting the matrix.

Potentially useful inverses of matrices:

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{pmatrix}^{-1} = \begin{pmatrix} 3 & -3 & 1 \\ -2.5 & 4 & -1.5 \\ 0.5 & -1 & 0.5 \end{pmatrix} \quad \begin{pmatrix} 5 & 6 & 14 \\ 6 & 15 & 36 \\ 14 & 36 & 99 \end{pmatrix}^{-1} = \begin{pmatrix} 0.3865 & -0.1840 & 0.0123 \\ -0.1840 & 0.6115 & -0.1963 \\ 0.0123 & -0.1963 & 0.0798 \end{pmatrix}$$

$$\begin{pmatrix} 4 & 6 & 14 \\ 6 & 14 & 36 \\ 14 & 36 & 98 \end{pmatrix}^{-1} = \begin{pmatrix} 0.95 & -1.05 & 0.25 \\ -1.05 & 2.45 & -0.75 \\ 0.25 & -0.75 & 0.25 \end{pmatrix}$$

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***Grading:***

- *(a) Design matrix – 5 pts*
- *(b) Ridge equation form – 4 pts*
- *(c) Numeric solution with explicit matrix steps – 9 pts*

*Total = 18 points*

## Solution

(a) **Design matrix $\mathbf{\Phi}$**

We have four data points, $x_1 = 0, x_2 = 1, x_3 = 2, x_4 = 3$, with corresponding targets $t_1 = 0, t_2 = 0.8, t_3 = 1.6, t_4 = 3.7$. Each row of $\mathbf{\Phi}$ is $(1,\ x_n,\ x_n^2)$:

$$\mathbf{\Phi} = \begin{pmatrix} 1 & 0 & 0^2 \\ 1 & 1 & 1^2 \\ 1 & 2 & 2^2 \\ 1 & 3 & 3^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{pmatrix}, \quad \mathbf{t} = \begin{pmatrix} 0 \\ 0.8 \\ 1.6 \\ 3.7 \end{pmatrix}.$$

(b) **Ridge equations**

We define $\lambda = 1$ and form

$$\big(\mathbf{\Phi}^\top \mathbf{\Phi} + \lambda\, \mathbf{I}\big)\, \mathbf{w} = \mathbf{\Phi}^\top \mathbf{t}.$$

We will solve for $\mathbf{w} = (w_0, w_1, w_2)^\top$.

(c) **Numeric solution**

$$\mathbf{\Phi}^\top\mathbf{\Phi} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 1 & 4 & 9 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{pmatrix} = \begin{pmatrix} 4 & 6 & 14 \\ 6 & 14 & 36 \\ 14 & 36 & 98 \end{pmatrix}$$

$$\mathbf{\Phi}^\top\mathbf{\Phi} + \lambda\mathbf{I} = \begin{pmatrix} 4 & 6 & 14 \\ 6 & 14 & 36 \\ 14 & 36 & 98 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 5 & 6 & 14 \\ 6 & 15 & 36 \\ 14 & 36 & 99 \end{pmatrix}.$$

$$\mathbf{\Phi}^\top\mathbf{t} = \begin{pmatrix} 1\cdot 0 + 1\cdot 0.8 + 1\cdot 1.6 + 1\cdot 3.7 \\ 0\cdot 0 + 1\cdot 0.8 + 2\cdot 1.6 + 3\cdot 3.7 \\ 0\cdot 0 + 1\cdot 0.8 + 4\cdot 1.6 + 9\cdot 3.7 \end{pmatrix} = \begin{pmatrix} 6.1 \\ 15.1 \\ 40.5 \end{pmatrix}.$$

$$(\mathbf{\Phi}^\top\mathbf{\Phi} + \mathbf{I})\,\mathbf{w} = \mathbf{\Phi}^\top\mathbf{t}.$$

$$\begin{pmatrix} 5 & 6 & 14 \\ 6 & 15 & 36 \\ 14 & 36 & 99 \end{pmatrix} \begin{pmatrix} w_0 \\ w_1 \\ w_2 \end{pmatrix} = \begin{pmatrix} 6.1 \\ 15.1 \\ 40.5 \end{pmatrix}.$$

$$(\mathbf{\Phi}^\top\mathbf{\Phi} + \mathbf{I})^{-1} = \begin{pmatrix} 5 & 6 & 14 \\ 6 & 15 & 36 \\ 14 & 36 & 99 \end{pmatrix}^{-1} = \begin{pmatrix} 0.3865 & -0.1840 & 0.0123 \\ -0.1840 & 0.6115 & -0.1963 \\ 0.0123 & -0.1963 & 0.0798 \end{pmatrix}.$$

$$\mathbf{w} = (\mathbf{\Phi}^\top\mathbf{\Phi} + \mathbf{I})^{-1} \begin{pmatrix} 6.1 \\ 15.1 \\ 40.5 \end{pmatrix} = \begin{pmatrix} 0.3865 & -0.1840 & 0.0123 \\ -0.1840 & 0.6115 & -0.1963 \\ 0.0123 & -0.1963 & 0.0798 \end{pmatrix} \begin{pmatrix} 6.1 \\ 15.1 \\ 40.5 \end{pmatrix} = \begin{pmatrix} 0.0774 \\ 0.1611 \\ 0.3428 \end{pmatrix}.$$

# Problem 2: Multiclass Logistic Regression with Three Classes (17 points)

**Statement.** We have three classes $C_1,\ C_2,\ C_3$ and one 2D data point *from each class*:

$$\mathbf{x}_1 = (0,1)\ \in\ C_1, \quad \mathbf{x}_2 = (1,2)\ \in\ C_2, \quad \mathbf{x}_3 = (2,2)\ \in\ C_3.$$

Our *multiclass* logistic model has three activation functions:

$$a_1(\mathbf{x}) = w_{1,0} + w_{1,1}\,x_1 + w_{1,2}\,x_2, \quad a_2(\mathbf{x}) = w_{2,0} + w_{2,1}\,x_1 + w_{2,2}\,x_2, \quad a_3(\mathbf{x}) = w_{3,0} + w_{3,1}\,x_1 + w_{3,2}\,x_2.$$

(a) Suppose we have the parameter vectors:

$$\mathbf{w}_1 = (-1,\ 1,\ 0), \quad \mathbf{w}_2 = (-1,\ 0,\ 1), \quad \mathbf{w}_3 = (0,\ -1,\ 2).$$

Compute $a_1(\mathbf{x}_n),\ a_2(\mathbf{x}_n),\ a_3(\mathbf{x}_n)$ *and* the $\mathrm{softmax}\big(a_1, a_2, a_3\big)$ for each data point $n = 1, 2, 3$.

(b) Which class is predicted for each of the three points?

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***Grading:***

- *(a) Softmax computations – 12 pts*
- *(b) Predicted class per point – 5 pts*

*Total = 17 points*

## Solution

(a) **Compute $a_k(\mathbf{x}_n)$ and the softmax for each point**

We are given:

$$\mathbf{w}_1 = (-1,\ 1,\ 0), \quad \mathbf{w}_2 = (-1,\ 0,\ 1), \quad \mathbf{w}_3 = (0,\ -1,\ 2).$$

Hence, for a point $\mathbf{x} = (x_1, x_2)$,

$$a_1(\mathbf{x}) = -1 + 1\cdot x_1 + 0\cdot x_2, \quad a_2(\mathbf{x}) = -1 + 0\cdot x_1 + 1\cdot x_2, \quad a_3(\mathbf{x}) = 0 + (-1)\cdot x_1 + 2\cdot x_2.$$

At $\mathbf{x}_1 = (0,1)$:

$$a_1 = -1 + 1\cdot 0 + 0\cdot 1 = -1, \quad a_2 = -1 + 1 = 0, \quad a_3 = 0 + (-1)\cdot 0 + 2\cdot 1 = 2.$$

$$p(1 \mid \mathbf{x}_1) = \frac{e^{-1}}{e^{-1} + 1 + e^2}, \quad p(2 \mid \mathbf{x}_1) = \frac{1}{e^{-1} + 1 + e^2}, \quad p(3 \mid \mathbf{x}_1) = \frac{e^2}{e^{-1} + 1 + e^2}.$$

At $\mathbf{x}_2 = (1,2)$:

$$a_1 = -1 + 1\cdot 1 + 0\cdot 2 = 0, \quad a_2 = -1 + 0\cdot 1 + 1\cdot 2 = 1, \quad a_3 = 0 + (-1)\cdot 1 + 2\cdot 2 = (-1+4) = 3.$$

$$p(1 \mid \mathbf{x}_2) = \frac{e^0}{1 + e + e^3}, \quad p(2 \mid \mathbf{x}_2) = \frac{e^1}{1 + e + e^3}, \quad p(3 \mid \mathbf{x}_2) = \frac{e^3}{1 + e + e^3}.$$

At $\mathbf{x}_3 = (2,2)$:

$$a_1 = -1 + 1\cdot 2 + 0\cdot 2 = 1, \quad a_2 = -1 + 0\cdot 2 + 1\cdot 2 = 1, \quad a_3 = 0 + (-1)\cdot 2 + 2\cdot 2 = 0 - 2 + 4 = 2.$$

$$p(1 \mid \mathbf{x}_3) = \frac{e^1}{2e^1 + e^2} = \frac{1}{2+e}, \quad p(2 \mid \mathbf{x}_3) = \frac{e^1}{2e^1 + e^2} = \frac{1}{2+e}, \quad p(3 \mid \mathbf{x}_3) = \frac{e^2}{2e^1 + e^2} = \frac{e}{2+e}.$$

(b) **Which class is predicted for each point?**

We pick the class $k$ whose $p(k \mid \mathbf{x}_n)$ is largest:

- For $\mathbf{x}_1 = (0, 1)$, $a_3 = 2$ is largest among $\{-1, 0, 2\}$, so $\exp(a_3) = e^2$ dominates. Likely class $C_3$ has the highest probability.
- For $\mathbf{x}_2 = (1, 2)$, $a_3 = 3$ is largest among $\{0, 1, 3\}$, so class $C_3$ again is highest.
- For $\mathbf{x}_3 = (2, 2)$, $a_3 = 2$ is largest among $\{1, 1, 2\}$, so once more class $C_3$ is highest.

# Problem 3: A Two-Layer Network with Sigmoid Hidden Units (17 points)

**Statement.** We have a neural network with a *single scalar input* $x \in \mathbb{R}$, two hidden units (each using a logistic sigmoid activation), and one linear output:

$$h_1 = \sigma(w_1\, x + b_1), \quad h_2 = \sigma(w_2\, x + b_2), \quad y = w_{o1}\, h_1 + w_{o2}\, h_2 + b_o,$$

where $\sigma(z) = \frac{1}{1+e^{-z}}$.

We have two training points:

$$(x_1, t_1) = (0,\ 1.0), \quad (x_2, t_2) = (2,\ 2.0),$$

and the error function is

$$E = \tfrac{1}{2} \sum_n \big(y_n - t_n\big)^2.$$

Suppose at initialization we set *all weights* $w_1, w_2, w_{o1}, w_{o2}, b_1, b_2$ to 0 except $b_o = 1$.

(a) Write down the chain-rule expressions for $\frac{\partial E}{\partial w_j}$ and $\frac{\partial E}{\partial b_j}$ at the hidden layer.

(b) Perform one gradient descent step with a small learning rate (say $\eta = 0.1$) on $(w_{o1}, w_{o2}, b_o)$, assuming $h_1 = h_2 = 0.5$. Show the updated values of $(w_{o1}, w_{o2}, b_o)$.

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***Grading:***

- *(a) Chain rule for hidden-layer parameters – 8 pts*
- *(b) Single gradient step on output layer – 9 pts*

*Total = 17 points*

## Solution

(a) **Chain-rule for hidden-layer parameters**

Let $z_j = w_j\, x + b_j$ and $h_j = \sigma(z_j)$. At the output we have $y = w_{o1}\, h_1 + w_{o2}\, h_2 + b_o$. Then

$$E = \tfrac{1}{2} \sum_n (y_n - t_n)^2, \quad (y_n - t_n) = \delta_n.$$

$$\frac{\partial E}{\partial w_j} = \sum_n \delta_n\, \frac{\partial y_n}{\partial w_j} = \sum_n \delta_n \left(\frac{\partial y_n}{\partial h_j}\right) \left(\frac{\partial h_j}{\partial w_j}\right),$$

$$\frac{\partial y_n}{\partial h_j} = w_{oj}, \quad \frac{\partial h_j}{\partial w_j} = \sigma'(z_j)\, x_n = h_j(1 - h_j)\, x_n.$$

$$\frac{\partial E}{\partial w_j} = \sum_n \delta_n\, w_{oj}\, h_j(1 - h_j)\, x_n.$$

$$\frac{\partial E}{\partial b_j} = \sum_n \delta_n\, w_{oj}\, h_j(1 - h_j).$$

(b) **Single gradient descent step on** $(w_{o1}, w_{o2}, b_o)$

At the start, $w_{o1} = w_{o2} = 0, b_o = 1$. We have $h_1 = h_2 = 0.5$ for each input. Let $y_n = w_{o1}\,0.5 + w_{o2}\,0.5 + 1 = 1$. Then:

$$\delta_n = (y_n - t_n).$$

For $(x_1 = 0, t_1 = 1)$ we get $\delta_1 = (1 - 1) = 0$. For $(x_2 = 2, t_2 = 2)$ we get $\delta_2 = (1 - 2) = -1$. Hence the partial derivatives for the output-layer weights:

$$\frac{\partial E}{\partial w_{o1}} = \sum_n \delta_n\, h_1^{(n)} = 0 \cdot 0.5 + (-1) \cdot 0.5 = -0.5,$$

$$\frac{\partial E}{\partial w_{o2}} = \sum_n \delta_n\, h_2^{(n)} = 0 \cdot 0.5 + (-1) \cdot 0.5 = -0.5,$$

$$\frac{\partial E}{\partial b_o} = \sum_n \delta_n = 0 + (-1) = -1.$$

With a learning rate $\eta = 0.1$, we update:

$$w_{o1} \leftarrow 0 - 0.1 \cdot (-0.5) = +0.05, \quad w_{o2} \leftarrow 0 - 0.1 \cdot (-0.5) = +0.05, \quad b_o \leftarrow 1 - 0.1 \cdot (-1) = 1.1.$$

So after one step: $(w_{o1},\, w_{o2},\, b_o) = (0.05,\, 0.05,\, 1.1)$.

# Problem 4: A Kernel Method with $\big(x\,x' + 2\big)^2$ (14 points)

**Statement.** We have three one-dimensional points:

$$x_1 = 1, \quad x_2 = 2, \quad x_3 = 3.$$

We define a polynomial kernel:

$$k(x, x') \;=\; \big(x\,x' + 2\big)^2.$$

(a) Compute the $3 \times 3$ Gram matrix $K_{ij} = k(x_i,\, x_j)$.

(b) Expand $\big(x\,x' + 2\big)^2$ to show the implied feature map $\phi(x)$.

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***Grading:***

- *(a) Gram matrix numeric – 8 pts*
- *(b) Feature-map expansion – 6 pts*

*Total: 14 points*

## Solution

(a) **Gram Matrix K**

We have three data points: $x_1 = 1, x_2 = 2, x_3 = 3$. The kernel is

$$k(x_i, x_j) = (\,x_i\,x_j + 2)^2.$$

Hence:

$$K_{11} = (1 \cdot 1 + 2)^2 = 3^2 = 9, \quad K_{12} = (1 \cdot 2 + 2)^2 = 4^2 = 16, \quad K_{13} = (1 \cdot 3 + 2)^2 = 5^2 = 25,$$

$$K_{22} = (2 \cdot 2 + 2)^2 = 6^2 = 36, \quad K_{23} = (2 \cdot 3 + 2)^2 = 8^2 = 64, \quad K_{33} = (3 \cdot 3 + 2)^2 = 11^2 = 121.$$

The full symmetric matrix is

$$\mathbf{K} = \begin{pmatrix} 9 & 16 & 25 \\ 16 & 36 & 64 \\ 25 & 64 & 121 \end{pmatrix}.$$

(b) **Feature-map expansion**

We note

$$(x\,x' + 2)^2 = x^2\,x'^2 + 4\,x\,x' + 4.$$

One valid feature map is (without worrying about constant scale factors):

$$\phi(x) = \big(x^2,\; 2x,\; 2\big).$$

Then $\phi(x)^\top \phi(x') = (x^2)(x'^2) + (2x)(2x') + (2)(2)$, which matches the above polynomial form.

# Problem 5: Hard-Margin SVM in 2D (14 points)

**Statement.** We have four points in $\mathbb{R}^2$, split into two linearly separable classes:

$$\text{Class } (-1)\colon (0,0),\ (1,0), \quad \text{Class } (+1)\colon (2,2),\ (3,3).$$

We want to:

(a) By inspection, propose a line that separates these classes.

(b) Identify the support vectors (the closest points to the decision boundary in each class).

(c) Compute the margin width of your line. Is this the maximum margin?

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***Grading:***

- *(a) Separating line – 5 pts*
- *(b) Support vectors – 6 pts*
- *(c) Margin width – 3 pts*

*Total = 14 points*

## Solution

(a) **Propose a separating line**

One simple choice is the line derived from the vector between points $(1,0)$ and $(2,2)$: $(1,2)$. This vector should serve as the normal to our decision boundary. Also, to maximize the margin this boundary should go through the center point between points $(1,0)$ and $(2,2)$: $(1,0) + 0.5(1,2)(1.5,1)$. Our implicit equation is:

$$f(\mathbf{x}) = x_1 + 2x_2 + c = 0 \quad \text{putting in point}(1.5,1) \quad f(\mathbf{x}) = x_1 + 2x_2 - 3.5 = 0.$$

(b) **Identify support vectors**

Points $(1,0)$ and $2,2$ are the support vectors as their value of $|f(\mathbf{x})|$ is the smallest.

(c) **Compute margin width**

As $|f((1,0))| = |f((2,2))|$ these two points are the support vectors and this line maximizes the margin. Margin width is half the length of the vector $(1,2)$: $\sqrt{5}/2$

# Problem 6: Gaussian Process (20 points)

**Statement.** We consider a *Gaussian Process* (GP) prior on a function $f(x)$ which has *zero mean* and uses the squared-exponential (also called RBF) kernel with amplitude $\alpha$:

$$k(x, x') \;=\; \alpha \, \exp\Big[-\tfrac{(x-x')^2}{2\,\sigma^2}\Big].$$

We have three noiseless observations:

$$(x_1,\, t_1) = (0,\, 0.5), \quad (x_2,\, t_2) = (1,\, 0.7), \quad (x_3,\, t_3) = (2,\, 1.2).$$

We want the *posterior* distribution of $f(x_*)$ for a new input $x_* = 1.5$.

(a) Construct the $3 \times 3$ kernel matrix $\mathbf{K}$ for the observed inputs $\{0, 1, 2\}$.

(b) Construct the 3-vector $\mathbf{k}_*$ which consists of $k(x_*, x_n)$ for $n = 1, 2, 3$.

(c) Write down the standard GP formulas for the predictive distribution and express mean and variance at $x_*$.

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*Grading:*

- *(a) Kernel matrix* $\mathbf{K}$ *– 7 pts*
- *(b) Vector* $\mathbf{k}_*$ *– 6 pts*
- *(c) Predictive distribution formulation mean and variance expressions – 7 pts*

*Total = 20 points*

## Solution

(a) **Kernel matrix $\mathbf{K}$**

Label the observed inputs as $x_1 = 0, x_2 = 1, x_3 = 2$. Then each $(i, j)$ entry is

$$K_{ij} = k(x_i, x_j) = \alpha \, \exp\Big[-\frac{(\,x_i - x_j\,)^2}{2\,\sigma^2}\Big].$$

Hence

$$\mathbf{K} = \begin{pmatrix} \alpha & \alpha\, e^{-\frac{(0-1)^2}{2\,\sigma^2}} & \alpha\, e^{-\frac{(0-2)^2}{2\,\sigma^2}} \\ \alpha\, e^{-\frac{(1-0)^2}{2\,\sigma^2}} & \alpha & \alpha\, e^{-\frac{(1-2)^2}{2\,\sigma^2}} \\ \alpha\, e^{-\frac{(2-0)^2}{2\,\sigma^2}} & \alpha\, e^{-\frac{(2-1)^2}{2\,\sigma^2}} & \alpha \end{pmatrix}.$$

(b) **Vector $\mathbf{k}_*$**

We want $k(x_*, x_n)$ for $x_* = 1.5$ and $n = 1, 2, 3$. Then

$$k(x_*, x_n) = \alpha \, \exp\Big[-\tfrac{(1.5-x_n)^2}{2\,\sigma^2}\Big].$$

So

$$\mathbf{k}_* = \begin{pmatrix} k(1.5, 0) \\ k(1.5, 1) \\ k(1.5, 2) \end{pmatrix} = \begin{pmatrix} \alpha \, \exp\big[-\frac{(1.5-0)^2}{2\,\sigma^2}\big] \\ \alpha \, \exp\big[-\frac{(1.5-1)^2}{2\,\sigma^2}\big] \\ \alpha \, \exp\big[-\frac{(1.5-2)^2}{2\,\sigma^2}\big] \end{pmatrix}.$$

(c) **Predictive distribution mean and variance at $x_*$:**

$$p(t_*|x_*,\ \mathbf{t},\ \mathbf{x})\ \sim\ \mathcal{N}\Big(\ \mathbf{k}_*^\top\,\mathbf{K}^{-1}\,\mathbf{t},\ k(x_*,x_*)\ -\ \mathbf{k}_*^\top\,\mathbf{K}^{-1}\,\mathbf{k}_*\Big),$$

where

$$k(x_*,x_*) = \alpha\ \exp\Big[-\tfrac{(1.5-1.5)^2}{2\,\sigma^2}\Big] = \alpha.$$

Therefore the predictive mean is

$$m_* = \mathbf{k}_*^\top\,\mathbf{K}^{-1}\,\mathbf{t}, \quad \text{and the predictive variance is} \quad \mathrm{Var}(f_*) = \alpha\ -\ \mathbf{k}_*^\top\,\mathbf{K}^{-1}\,\mathbf{k}_*.$$